Landau theoretical minimum: PFP II

AuthorityEffect: None. The Apartment 50 preface is fictional examination staging. Use the study guide to prepare and keep the separate instructor key closed during an attempt. The manuscript defines the claims.

This is a written and oral problem set, not a historical Landau examination. Show equations, witnesses, and the scope of each conclusion. Use exact integers and fractions; numerical plots or a solver's verdict alone do not replace the requested argument.

Common instance and conventions

For Q1-Q4 and Q8-Q10, the ordered vertices are `(a,b,c)` and the undirected edges are indexed as follows:

Edge ID	Endpoints	Parity bit
0	a, b	1
1	b, c	1
2	c, a	0

The gauge equation is (_u __v=c__e). A walk step is `(edge_id, departure, arrival)`. Multiple edges and self-loops are allowed. An arrow in a walk records traversal; it does not make the edge directed or attach a face.

The search uses all (2^3=8) assignments in zero-based lexicographic order: `000,001,010,011,100,101,110,111`. Intervals are half-open: ([s,t)) includes (s) and excludes (t). Root-fixing in a proof does not change this implemented search space.

Level 0: Declare the question

Q1. Carrier and move grammar (6 points). Specify the carrier, coefficient system, allowed transformation, and acceptance equation for the common instance. What do bits zero and one require? Explain why a vertex name, a bracket-wire encoding, or an unfilled graph drawing supplies no extra mathematical structure.

Level 1: Produce and attack a witness

Q2. Gauge and its freedom (6 points). Find a gauge with (_a=0) and check all three equations. Find every gauge without that root condition. Explain what changes if a second connected component is added.

Q3. Three minimal failures (6 points). Change edge 2 to parity one. Give an indexed odd closed walk and prove that a gauge is impossible. Then give

an obstruction on one vertex and on two vertices joined by two parallel edges. State the parity bits in each case. Explain why forbidding loops would change the declared input grammar.

Level 2: Prove the criterion and respect coefficients

Q4. Same displayed numbers, different coefficients (8 points). Instead interpret the reports ((1,1,0)) as rational numbers on the directed cycle (a→b→c→a), with an exact potential requiring (r_{uv}=p_v-p_u). Compute the rational cycle sum and the binary parity. Does a rational potential exist? Find the minimum uniform (^1) change to the three reports needed to make them exact, with both a lower bound and an attaining repair. Do not import a filled triangle.

Q5. Both directions of balance (10 points). Prove that a finite graph admits a gauge exactly when every closed walk has parity zero. Handle disconnected components, loops, and parallel edges. Explain how a spanning forest produces either witness in (O(|V|+|E|)), and why that algorithm is the appropriate baseline for this grammar.

Level 3: Transfer the witness to an exact matrix

Q6. Work the adapter (8 points). For

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \qquad D = \operatorname{diag}(1, -1, 1),$$

derive the three graph bits and compute (DHD). Explain how zero off-diagonal entries are treated. Give the input-contract response to a float entry, a nonsquare matrix, and a nonsymmetric matrix. Does the all-positive triangle obstruct an unrestricted global eigenbasis?

Q7. Positivity without an efficiency claim (10 points). Let (H') be a finite real symmetric stoquastic matrix. For (0), prove that (e^{- H'}) is entrywise nonnegative by choosing (B=aI-H'). State why the series converges, what happens at (=0), and what additional evidence is needed before claiming efficient sampling or low-variance estimates for observables.

Level 4: Account for every tested candidate

Q8. Exact frontier trace (8 points). The common instance receives budget two from cursor zero. List the tested assignments and their verdicts, the receipt status, the witness field, and the remaining interval. Resume with budget one and the first receipt. Give the new checks, status, witness, and cursor. What does checking that second receipt require besides its own digest?

Q9. Adversarial continuation (6 points). Explain the result of each attempt: starting at index two with no prior receipts; flipping a recorded verdict

and recomputing its digest; reusing an old receipt after changing the declared basis while retaining the Hamiltonian bytes. Name the evidence that makes a valid continuation possible. Explain why an exact retrieval key still does not certify a retrieved gauge or a physical adapter.

Q10. Batches are part of the trace (6 points). Run the common instance from zero with budget eight. The implementation checks fixed batches of at most eight and finishes a batch before stopping. Which witness is selected, how many checks are recorded, and what is the cursor? Would workers one and four change these answers? Contrast this with Q8 and state what a zero-budget initial call establishes.

Level 5: Check cancellation, not just the total

Q11. Pairing obligations (4 points). State the four requirements for a sign-reversing equal-magnitude pairing on its paired subset. Explain why the carrier with contributions (+2,-1,-1) has nonnegative total but no positive equal-magnitude partner for either negative term. Does allowing a collective regrouping ask the same question? Why can an absolutely convergent thermal expansion of a nonzero finite-dimensional Hermitian system not have every term canceled in such pairs?

Level 6: Close the unbounded length argument

Q12. Derive the target weight (8 points). Use the six-mode order, parameters, and hopping-sign rule of manuscript Section 6 for

$$110;111 \longrightarrow 101;111 \longrightarrow 011;111 \longrightarrow 110;111.$$

Compute the three hopping signs, each visited energy, the simplex volume, and the signed integrated weight. Explain why 64 occupation states do not make the carrier of all rooted closed hopping sequences finite.

Q13. Exclude a partner at every length (8 points). Use the exact finite facts: 192 rooted three-hop sequences, 96 positive and 96 negative; positive three-hop paths visit energies at most two; all energies lie in ([0,6]); hops have magnitude one; and no one-hop closure exists. Derive the bounds for lengths (0,1,2,3, 4) and the strict gap around (W_ =e^{-1}/384). Explain how alternating rational partial sums certify the comparisons, why the long-path bound decreases, and why it bounds one path rather than the total tail.

Level 7: Defend the publication claim

Q14. Evidence audit (6 points). Write a short audit with one exact claim, its declared domain, a relevant package test or checker, and the limit of that evidence. Address these proposed conclusions:

- "Two failures prove there is no gauge."

- "The path gap proves that this Hamiltonian has no sign-free representation."
- "The manifest verifies the mathematics."
- "A Python fallback test verifies the native executable."

State the narrower conclusion, missing evidence, or refusal in each case. Do not require a general physical adapter as a prerequisite for the already scoped mathematical result.

Scoring and oral defense

The points sum to 100. The key allocates partial credit to calculations and arguments; a bare correct label receives no calculation or proof credit. A score of at least 80 with no unresolved scope error is the instructional competence target, not a validated learner metric.

After marking, defend three answers orally: Q5's sufficiency proof, Q8's continuation, and Q13's all-length gap. Correct any promotion of UNKNOWN to an obstruction, of a digest to truth, or of a restricted obstruction to a general physical classification before recording mastery.